PROGRAMMING FUNDAMENTALS – FINAL EXAM PRACTICE SHEET

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Built to practice logic, arrays, sorting, and patterns – All in one place!

LEVEL 0 – WARM UP (Basics Recap)

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1. Take your name and age as input. Print a greeting.
2. Add 3 numbers and print their sum.
3. Swap two numbers using a third variable.
4. Swap two numbers without using a third variable.
5. Write a calculator program that takes two numbers and a symbol (+, -, *, /, %) and performs the operation.

LEVEL 1 – LOOPING LOGIC

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6. Print numbers from 1 to N (try all 3 loops).
7. Print numbers from N to 1.
8. Print the sum of first N natural numbers.
9. Print the product of numbers from 1 to N (Factorial).
10. Count how many digits are in a number.
11. Find the sum of digits of a number.
12. Reverse the digits of a number.
13. Check if a number is a palindrome.
14. Check if a number is an Armstrong number.
15. Generate first N terms of the Fibonacci sequence.

LEVEL 2 – PATTERN PRINTING MASTER

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16. Print a solid square of `*`.
17. Print a right-angled triangle of `*`.
18. Print a number triangle like:

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1
12
123
1234
```

19. Print Floyd’s triangle:

1
2 3
4 5 6
7 8 9 10

20. Print a full pyramid of `*`.
21. Print Pascal's triangle.
22. Print an alphabet triangle (A, AB, ABC…).
23. Print a hollow square pattern.
24. Print a checkerboard pattern using `#`.

LEVEL 3 – ARRAY ESSENTIALS (1D Arrays)

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25. Take N elements in an array. Print all elements.
26. Find the sum of all array elements.
27. Find the maximum and minimum element.
28. Count even and odd elements in the array.
29. Search for a number in the array.
30. Reverse the array.
31. Copy one array into another.
32. Count how many times a number appears in the array.
33. Find the second largest number.
34. Print only unique elements in the array (no repeats).

LEVEL 4 – ARRAY + LOGIC COMBO

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35. Find the element that appears only once (others appear twice).
36. Remove duplicates from an array.
37. Count how many elements are greater than average.
38. Shift array elements to the left (rotation).
39. Shift array elements to the right (rotation).
40. Merge two arrays into one.
41. Find common elements between two arrays.
42. Replace every element with its square.
43. Find the frequency of each number in the array.
44. Count how many pairs have a sum equal to a target number.
45. Check if an array is a palindrome.

LEVEL 5 – SORTING & MORE LOGIC

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46. Sort an array using Bubble Sort.
47. Sort an array using Selection Sort.
48. Sort an array using Insertion Sort.
49. Sort array in descending order.
50. Sort even numbers first, then odd numbers (same array).
51. Sort a string alphabetically.

52. Sort an array of names alphabetically.
53. After sorting, search for a number using Binary Search.
54. Merge two sorted arrays into one sorted array.
55. Sort array and remove duplicates.

LEVEL 6 – FINAL COMBO ROUND – HARDEST QUESTIONS

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56. Remove a specific number from the array (shift remaining).
57. Rearrange array to alternate positive & negative numbers.
58. Print leaders in an array (element greater than all right ones).
59. Check if the array is sorted or not.
60. Find the difference between max and min in array.
61. Sum all the digits of all elements in the array.
62. Convert an array of digits to a number.
63. Find all duplicate elements and how many times they appear.
64. Print frequency of each digit in a number.
65. Build a mini calculator using switch (menu driven).

TIPS TO SCORE FULL MARKS – FINAL EXAM

Just a few things that are really helping me while practicing PF, thought I'd share:

- Try to practice daily, even if it's just 3–5 questions at night — it really adds up over time.
- Don't just focus on the syntax. Try to understand *why* each step works — that's where the real learning happens.
- Dry run your code on paper, especially for loops and tricky index stuff — it makes things way clearer.
- Keep your code neat. Use proper variable names so it's easier to read and debug later.
- Don't skip the tough questions. They might take time, but they build confidence super fast.
- Try to guess the output before running your code — great way to improve debugging skills.
- And yeah, push your work to GitHub if you can. It's a good way to track your progress and revisit stuff later.

END OF SHEET – MASTER THIS = YOU'RE UNSTOPPABLE
